

Conservation, Forcing, and Speedup in Reverse Math

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Quick Review

Recall

L_2 is the language of second order arithmetic.

L_2 structures have two sorts – “numbers” and “sets”.

The intended model is $(\omega, \mathcal{P}(\omega))$.

An L_2 struture (ω, S) with $S \subseteq \mathcal{P}(\omega)$ is called an ω -model.

Recall

Basic = first order axioms of arithmetic without induction

RCA_0 = Basic + Σ_1^0 -induction + Δ_1^0 -comprehension

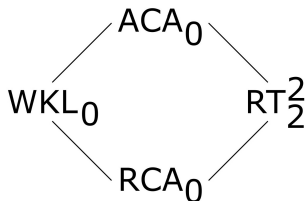
WKL = Weak König's Lemma: All infinite binary trees have paths

WKL_0 = $\text{RCA}_0 + \text{WKL}$

RT_2^2 = Ramsey's Theorem for pairs: Every two-coloring of unordered pairs has a monochromatic set

Recall

We discussed relative strength using RCA_0 as the base theory.
“Stronger” here meaning “proves the same and more.”



How much stronger are WKL_0 and RT_2^2 than RCA_0 ?
What does “how much stronger” even mean?

Harrington's Result

Conservation

Strength =? what kind of new things can be proven

Result. [Harrington '77] WKL_0 is Π_1^1 -conservative over RCA_0

That is – $\text{WKL}_0 \vdash \phi$ if and only if $\text{RCA}_0 \vdash \phi$,
when ϕ can be written in the form $\forall X, \psi(X)$
with ψ arithmetic.

In general: Given theories $T \subseteq T'$ and a collection of sentences Γ ,
we say that T' is Γ -conservative over T if,
for all $\psi \in \Gamma$, $T' \vdash \psi$ if and only if $T \vdash \psi$.

Proving Π_1^1 Conservation

Define: Given an L_2 -structure $\mathcal{M} = (M, S_M)$, for any $S_N \subseteq S_M$, the structure $\mathcal{N} = (M, S_N)$ is called an ω -submodel of $\mathcal{M}$, denoted $\mathcal{N} \subseteq_\omega \mathcal{M}$.

Note: An ω -submodel need not be an ω -model, but every ω -model is an ω -submodel of $(\omega, \mathcal{P}(\omega))$.

Lemma. It suffices to show that every countable model of RCA_0 is an ω -submodel of some model of WKL_0 .

Why? If $\phi \in \Pi_1^1$ is not proven by RCA_0 , then it is false in some model $\mathcal{M}$. By Lowenheim-Skolem, we may assume $\mathcal{M}$ is countable. Since $\mathcal{M} \models \neg\phi$, there is some $A \in \mathcal{M}$ witnessing this. But if $\mathcal{M} \subseteq_\omega \mathcal{N}$ for $\mathcal{N} \models \text{WKL}_0$, $A \in \mathcal{N}$. So $\text{WKL}_0 \not\models \phi$.

Harrington

Lemma. For all countable $\mathcal{M} \models \text{RCA}_0$, there exists $\mathcal{N} \models \text{WKL}_0$ such that $\mathcal{M} \subseteq_\omega \mathcal{N}$.

The proof progresses in stages:

- force a generic path G for some binary tree T
- show G preserves Σ_1^0 -induction
- expand with new Δ_1^0 -definable subsets
- repeat to form a chain

Forcing the Path

Let $\mathcal{M} = (M, S_m)$ be a countable model of RCA_0 .

Forcing poset: the collection of $\mathcal{M}$ -definable infinite binary trees ordered by the subtree relation.

Dense subsets: all $\mathcal{M}$ -definable dense subsets.

Given an $\mathcal{M}$ -definable infinite binary tree T , the generic filter G is a descending chain of subtrees of T , so we can interpret G as its intersection.

“this tree has only one node at level i ” is one of our dense subsets, so G is a path for T .

Preserving Induction

Let $\mathcal{M}' = (M, S_M \cup \{G\})$.

Lemma. For every Σ_1^0 formula $\phi(n, G)$ definable in $\mathcal{M}$, there exists a normal form $\exists z, \theta(n, G \upharpoonright z)$, where θ is Δ_0^0 .

Why? Think of this as the proof-theoretic reflection of the Finite Use Principle for Turing machines using oracle G . Our $\phi(n, G)$ has form $\exists x, \psi(x, n, G)$. We recursively translate this in such a way to reflect that we can bound both x and the initial segment of G we need by some z .

Preserving Induction

Using this normal form $\exists z, \theta(n, G \upharpoonright z)$:

Define. For $a \in M$, define D_a to be the collection of $\mathcal{M}$ -definable trees T such that, for all $n \leq_M a$, either:

- $\forall \sigma \in T, \neg \theta(n, \sigma)$, or
- $\exists k, \forall \sigma \in T \cap 2^k, \exists \tau \preceq \sigma, \theta(n, \tau)$.

Lemma. For all $a \in M$, D_a is $\mathcal{M}$ -definable and dense.

Why? Note that for $T \in D_a$, what the definition is saying is that, for all $n \leq_M a$, either $\neg \phi(n, G)$ for all possible paths G through T , or $\phi(n, G)$ for all paths. To show density is to show that we can trim a tree so that its possible paths will all agree on ϕ at n .

Preserving Induction

Lemma. If G is generic, each set $I_a = \{n \leq_M a \mid \mathcal{M}' \models \neg\theta(n, G)\}$ is finite.

Why? This is just a technical argument about induction, bounding, and what “finite” means in RCA_0 .

If $\forall n, \phi(n, G)$ does not hold, there is some $a \in M$ such that $\neg\phi(a, G)$. So $a \in I_a$ and I_a is finite, so there is a least n for which $\mathcal{M}' \models \neg\phi(n, G)$.

So we can conclude that $\mathcal{M}' \models \text{Basic} + \Sigma_1^0\text{-induction}$.

Add Δ_1^0 -definable

We still need to satisfy Δ_1^0 -comprehension.

There are only countably many sets Δ_1^0 -definable in $\mathcal{M}'$. Since Δ_1^0 sets have a Σ_1^0 form, adding these will not break induction.

Let S_G be the union of these with $S_M \cup \{G\}$ and $\mathcal{M}[G] = (M, S_G)$.

So we have:

- $\mathcal{M} \subseteq_\omega \mathcal{M}[G]$
- $\mathcal{M}[G]$ is countable
- $\mathcal{M}[G] \models \text{RCA}_0$
- for one of the countably many trees definable in $\mathcal{M}$, there is a path in $\mathcal{M}[G]$

Form Chain

We want a chain

$$\mathcal{M} \subseteq_{\omega} \mathcal{M}[G_1] \subseteq_{\omega} \mathcal{M}[G_1][G_2] \subseteq_{\omega} \cdots$$

At each stage up to countably many more definable trees are added, but with care we can construct the chain so they all receive a path at some later stage.

Let $\overline{\mathcal{M}}$ be the union over this chain.

Then $\overline{\mathcal{M}} \models \text{WKL}_0$ and $\mathcal{M} \subseteq_{\omega} \overline{\mathcal{M}}$.

Sufficient to prove Π_1^1 -conservation by previous argument.

Avigad's Technique

Forcing in RCA_0

The formal forcing argument proceeds roughly the same:

- force a generic path G for some binary tree T
- show G preserves Σ_1^0 -induction
- expand with new Δ_1^0 -definable subsets
- repeat to form a chain (kind of...)

Forcing the Path

Define $Condition(P)$ to be the formula stating “ P is an infinite binary tree” and $Q \leq P$ by inclusion.

RCA_0 proves our forcing conditions form a poset.

The name for set X is:

- $\{\langle 0, x \rangle \mid x \in X\}$ for each X in the base model,
- $\{\langle 1, 0 \rangle\}$ if X is the generic G .

Define $Name(X)$ to be the formula stating that X has one of these two forms.

Forcing the Path

For atomic ϕ , we define $P \Vdash \phi$ by cases:

- if ϕ is $t_1 = t_2$, $t \in X$, etc., let $P \Vdash \phi$ be simply ϕ
- if ϕ is $t \in G$, let $P \Vdash \phi$ say that every path of P contains t (we can do this Σ_1^0)

Lemma. This forcing notion satisfies the following in RCA_0 :

- Monotonicity: for conditions P and Q , if $Q \leq P$ and $P \Vdash \phi$, then $Q \Vdash \phi$; similarly for $P \Vdash \text{Name}(X)$ implying $Q \Vdash \text{Name}(X)$
- Substitution: $P \Vdash (\phi[x, t])$ if and only if $(P \Vdash \phi)[x/t]$
- Weakness: $P \Vdash \phi$ if and only if $P \Vdash \neg\neg\phi$

(This is what Avigad calls a good weak forcing notion.)

Forcing the Path

This can be extended to arbitrary formulas. We define $P \Vdash \phi$ by cases of ϕ :

- $\neg\psi$: $\forall Q \leq P, (Q \nVdash \psi)$
- $\psi_1 \wedge \psi_2$: $(P \Vdash \psi_1) \wedge (P \Vdash \psi_2)$
- $\psi_1 \vee \psi_2$: $\forall Q \leq P \exists R \leq Q, ((R \Vdash \psi_1) \vee (R \Vdash \psi_2))$
- $\psi_1 \rightarrow \psi_2$: $\forall Q \leq P, (Q \Vdash \psi_1 \rightarrow \exists R \leq Q (R \Vdash \psi_2))$
- $\exists n, \psi(n)$: $\forall Q \leq P \exists R \leq Q, \exists n (R \Vdash \psi(n))$
- $\forall n, \psi(n)$: $\forall n (P \Vdash \psi(n))$
- $\exists X, \psi(X)$: $\forall Q \leq P \exists R \leq Q, \exists X (R \Vdash \text{Name}(X) \wedge \psi(X))$
- $\forall X, \psi(X)$: $\forall X (P \Vdash \text{Name}(X) \rightarrow \psi(X))$

Lemma. In general, extending a good weak forcing notion in this way preserves monotonicity, substitution, and weakness.

Forcing the Path

Lemma. For arithmetic $\phi \in \Sigma_1^0$ with parameters $X_1, \dots, X_n$ in the base model, RCA_0 proves TFAE:

- $\forall P, (P \Vdash \phi(X_1, \dots, X_n))$
- $\exists P, (P \Vdash \phi(X_2, \dots, X_n))$
- $\phi(X_1, \dots, X_n)$

$\text{RCA}_0 \vdash (P \Vdash \phi)$ can be taken to roughly mean: $\mathcal{M}' \models \phi$, where $\mathcal{M}' = (M, S_M \cup \{G\})$ and G is a generic path through P .

By the above lemma, RCA_0 proves that if we extend a model in this way, it will satisfy the Basic axioms.

But most importantly, RCA_0 proves $P \Vdash (G \text{ is a path through } P)$.

Preserving Induction

This proceeds similarly to the same step in Harrington's proof:

- use the normal form of Σ_1^0 formulas
- use the properties of finiteness in RCA_0
- expand the definitions of forcing

Lemma. If ϕ is Σ_1^0 , then $P \Vdash \phi$ can be written in Σ_1^0 form.

Conclusion: Σ_1^0 -induction holds for the forcing notion.

Add Δ_1^0 -definable

For this we need another forcing notion ($\Vdash_1$).

- use the same forcing poset as previously
- we name all sets Δ_1^0 -definable with a parameter X from the base model and generic G by triples $\langle X, \psi_1, \psi_2 \rangle$ such that X is the parameter from the base model and ψ_1, ψ_2 are Σ_1^0 and Π_1^0 definitions of the Δ_1^0 set.
- $P \Vdash_1 \text{Name}(X)$ is defined in terms of P forcing (by the previous notion) the Σ_1^0 and Π_1^0 definitions to be equivalent
- $P \Vdash t \in X$ is defined in terms of P forcing (again, previous notion) t to satisfy one of the definitions

This is a good weak forcing notion.

Add Δ_1^0 -definable

Similar to the previous forcing notion,

- this extends to a good weak forcing notion on arbitrary formula
- RCA_0 proves arithmetical ϕ if and only if RCA_0 proves some condition (equivalently, all conditions) force ϕ
- we still preserve Σ_1^0 -induction

Most importantly, we now satisfy Δ_1^0 -comprehension

So to compare with the Harrington proof: $\text{RCA}_0 \vdash (P \Vdash_1 \phi)$ means roughly $\mathcal{M}[G] \models \phi$.

Form Chain??

Here is where problems arise: We need to repeat this process an infinite number of times, but then our forcing definitions would be infinitely long.

What we need instead is a new forcing notion $\Vdash_n$ that uniformly provides the results that repeating the previous n times would.

We can primitive recursively define the notions of $Condition_n(P)$, $Q \leq_n P$, $Name_n(X)$, and $t \in_n X$, allowing us to define $\Vdash_n$ uniformly in n . That is, we have relatively well-bounded algorithm that takes n and gives us the relevant forcing definition.

Two things to note:

- n -conditions basically encode sequences of trees
- taking the sequences to be in reverse, $Q \leq_n P$ is equivalent to $\bigwedge_i Q \restriction (n-i) \Vdash_{n-i} [Q(n-i) \Vdash_1 P(n-i)]$

Form Chain??

But $\Vdash_n$ still does not get us an entire chain.

We need one more forcing notion $\Vdash_\omega$.
(I promise, this is the last one.)

Our forcing conditions are all of the n -conditions ordered by $\leq_n$ and reverse ordering on n . (Explain.)

$P \Vdash_\omega \text{Name}(X)$ and $P \Vdash_\omega t \in X$ are defined hierarchically – once “we get to a level k where all conditions and names are defined, it must be k -forced by P .”

We obtain a good weak forcing notion on arbitrary ϕ that carries useful properties of the previous notions – most importantly, $(P \Vdash \phi) \rightarrow \phi$ for arithmetical sentences.

Form Chain??

At this step of the Harrington proof, we pointed out that we needed to take care in the order we added generics.

Here, we are basically forcing $\Vdash_n$ for all n .
(Technically, for all n in a definable cut.)

Generics paths are added generically, order isn't a concern.

Roughly, $P \Vdash_\omega \phi$ is equivalent to $\overline{M} \models \phi$ in the Harrington proof.

Form Chain??

The arithmetic statements of the base model (and so Basic) were preserved since $\Vdash_\omega$ is equivalent to an infinite repetition of $\Vdash_1$.

The same for Σ_1^0 -induction and Δ_1^0 -comprehension.

Now, though:

If $P \Vdash_\omega \text{Name}(X) \wedge \text{Tree}(X)$, RCA_0 proves that there is a $Q \leq_\omega P$ s.t. $Q \Vdash_\omega \exists Y (Y \text{ is a path through } X)$.

In some sense, we are approximating WKL.

The Result

If $WKL \vdash \forall X, \phi(X)$ with ϕ arithmetic, there exists P s.t.
 $RCA_0 \vdash (P \Vdash_\omega \forall X, \phi(X))$.

By definition, $RCA_0 \vdash \forall X, (P \Vdash_\omega \phi(X))$, so for any A ,
 $RCA_0 \vdash (P \Vdash_\omega \phi(A))$.

But by equivalence results, this means that for any A ,
 $RCA_0 \vdash \phi(A)$.

Hence $RCA_0 \vdash \forall A, \phi(A)$, demonstrating Π^1_1 -conservation.

Proof Length

Suppose $\text{WKL}_0 \vdash \forall X, \psi(X)$ for ψ arithmetic.

Let $\Gamma = (\phi_0, \dots, \phi_n = \forall X, \psi(X))$ be a proof of $\forall X, \psi(X)$ in WKL_0 – that is, for each i , ϕ_i is either an axiom, or follows from $\Gamma \upharpoonright i$ by inference.

The definitions of $\Vdash_\omega$ allow us to map each element ϕ_i to a forcing statement.

Since $P \Vdash_\omega (\forall X, \psi(X))$ implies $\forall X, \psi(X)$ in RCA_0 , we can use Γ to construct a proof Γ' in RCA_0 .

The details of the $\Vdash_\omega$ definitions establish that the number of symbols in each forcing statement is polynomial to the number of symbols in the corresponding proof step. So the length of Γ' is not significantly longer than Γ .

Recent Work

Proof Speedup

It's a straightforward step to move from speaking about the relative length of proofs to proof speedup.

If our translation occurs in polynomial time, this also implies that it has no more than a polynomial increase in length.

So rather than compare the length of proofs, we can compare the speed of generating the proofs.

Proof Speedup

Avigad's method was recently used to strengthen the following:

[Patey, Yokoyama '18] $WKL_0 + RT_2^2$ is $\forall\Sigma_2^0$ -conservative over RCA_0 .

That is, $WKL_0 + RT_2^2$ proves no new statements that universally quantify (first or second order) over Σ_2^0 formulas.

[Kołodziejczyk, Wong, Yokoyama '21] $WKL_0 + RT_2^2$ has only polynomial speedup over RCA_0 , but non-elementary speedup over RCA_0^* , for $\forall\Sigma_2^0$ statements.

Here the authors used a good strong forcing notion (as opposed to the good weak notion we used).

Proof Speedup

The authors restate Avigad's general result:

Result. Let T, T' be theories and Γ a collection of sentences in their common language. If there is a forcing interpretation of T' in T s.t. for all $\phi \in \Gamma$, $T \vdash [\forall P, (P \Vdash \phi)] \rightarrow \phi$, then T' is Γ -conservative over T . Moreover, if the interpretation is polynomial, and, given ϕ , a proof of $[\forall P, (P \Vdash \phi)] \rightarrow \phi$ can be found in polynomial time, then T' has at best polynomial speedup over T for Γ .

What I'm Up To

WKL and RT_2^2 are both combinatorial statements.

In the first talk we established that both were strictly between RCA_0 and ACA_0 in implicational strength.

The results today establish some upper bounds on their strength with conservation and proof complexity results.

It's natural to ask similar questions about other combinatorial statements.

Milliken's Tree Theorem

Define a tree T in this sense as a subset of $\omega^{<\omega}$ for which:

- Rooted: there exists a element ρ s.t. for all $\omega \in T$, $\rho \preceq \omega$
- Meet closed: for $\sigma, \tau \in T$, the longest common initial segment is also in T
- Finitely branching: each $\sigma \in T$ has a finite number of successors

Note: This is different from the closed-downward trees we usually use in computability!

The **level** of a node in T is just the count of how many of its predecessors are in T .

Milliken's Tree Theorem

A **Strong Subtree** of height α of T is a subtree of T with maximum level α that preserves the relative levels of nodes from T , and preserves the number of branchings for each node.

In a sense, a strong subtree of height α is isomorphic to T restricted to α .

Milliken's Tree Theorem. Let T be an infinite tree with no leaves. For all k -colorings of strong subtrees of T with height n , there is a strong subtree of T with height ω which is monochromatic.

Take MTT^n to be the formal statement in L_2 of the restriction of Milliken's to k -colorings on subtrees of height n .

Milliken's Tree Theorem

Fact. MTT^2 proves RT_2^2 .

Why? Use the coloring of pairs to define a coloring on pairs of levels in T .

[d'Auriac, Cholak, Dzhafarov, Monin, Patey '21]
proved the following:

Theorem. $\text{ACA}_0 \vdash \text{MTT}^n$.

Theorem. MTT^2 is cone avoiding.

So MTT^2 is strictly weaker than ACA_0 .

The Question

Since MTT^2 is related to RT_2^2 , can we get conservation and speedup results through similar methods?